KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 18 · MATH

The Capital Street Survey — Teacher Lesson Plan & Answer Key

Learning objectives: Students identify points, lines, line segments, rays, and parallel and perpendicular lines; measure and draw angles with a protractor and classify them as acute, right, obtuse or straight; find angles as turns of a 360° circle (including clock angles); and use angle addition to find missing angles in 90°, 180° and 360° wholes.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Imani, who helps her uncle Mr. Adeyemi, a land surveyor, and Ms. Ruiz, a bike-lane planner, measure street corners in Washington, D.C. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Dividing Decimals Review (4 pts, 2 each, Week 17 skill).
3. Slide 2 — Warm-Up, Part 2: Estimate and Solve (4 pts, 2 each): estimate first to place the decimal point.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W18.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching. A protractor helps on this sheet.
5. Slide 4 — A City Planned with Lines and Angles: nonfiction connection about Pierre L'Enfant's 1791 plan for Washington, D.C., with a grid and diagonal avenues. No questions.
6. Slide 5 — Teach: Points, Lines, Segments, and Rays. Have the student trace each with a finger: a line has two arrows, a ray one, a segment none.
7. Slide 6 — Your Turn: Lines and Rays (8 pts, 4 each).
8. Slide 7 — Your Turn: Counting Right Angles (4 pts): sketch the streets, count the crossings, then the corners at each crossing.
9. Slide 8 — Teach: Angles and the Protractor. Mr. Adeyemi's corners: put the center hole on the vertex, line one ray up with 0°, and pick the scale that starts at 0. Check: is the angle acute or obtuse before reading?
10. Slide 9 — Your Turn: Measure and Sort Angles (8 pts, 4 each).
11. Slide 10 — Your Turn: Clock Angles (4 pts): 12 equal spaces of 30° each.
12. Slide 11 — Teach: Angle Addition and Missing Angles. The whole equals the sum of the parts; know the whole first (90°, 180° or 360°).
13. Slide 12 — Your Turn: Angle Addition (8 pts, 4 each).
14. Slide 13 — Your Turn: More Missing Angles (4 pts).
15. Slide 14 — Vocabulary Round-Up: 10 flip cards (point, line segment, ray, parallel lines, perpendicular lines, vertex, degree, protractor, acute angle, obtuse angle).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.

17. Slide 16 — Teach: Drawing Angles and Angles as Turns. Ms. Ruiz's 65° bike lane drawn with a protractor; a 1/4 turn is 90°, a 1/2 turn is 180°.
18. Slide 17 — Your Turn: Turns and Drawing (6 pts, 3 each).
19. Slide 18 — Real World: Ms. Ruiz's Bike Lanes (4 pts): a different angle situation in each set (avenue and street, plaza paths, crosswalk ramp).
20. Slide 19 — Real World: The Surveying Tool (4 pts): equal clicks of 15°, 12° or 20°; multiply or divide to find the turn.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): find the subject before choosing the verb ("of the streets" can hide it). Same exact-check rules; where shown, both serial-comma forms or either verb tense are accepted.
24. Slide 23 — Match-Up Round (10 pts): Set 1 matches angle names to measures, Set 2 matches figures to descriptions, Set 3 matches fractions of a turn to degrees; 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Big Intersection (8 pts, 4 each): two multi-step angle questions about a street corner, a traffic circle, or paths along a curb.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W18.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Street Survey Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Capital City Angle Master!" with the congratulations message "You finished the street survey of Washington, D.C.!" and a big confetti burst; below 85 shows "Good Work on the Streets!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: DIVIDING DECIMALS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Dividing a decimal by a whole number (Week 17): type the quotient. 9.2 ÷ 4 =	**2.3**	9.2 ÷ 4: 8 ÷ 4 = 2, and the 1.2 left over ÷ 4 = 0.3, so the quotient is 2.3. Check: 2.3 × 4 = 9.2.
r2_s0	Set 1: Dividing by 10, 100, or 1,000 (Week 17): type the quotient. 36.5 ÷ 100 =	**0.365** (also accepted: .365)	Dividing by 100 moves every digit two places to the right (to smaller place values): 36.5 ÷ 100 = 0.365.
r1_s1	Set 2: Dividing a decimal by a whole number (Week 17): type the quotient. 9.6 ÷ 3 =	**3.2**	9.6 ÷ 3: 9 ÷ 3 = 3 and 0.6 ÷ 3 = 0.2, so the quotient is 3.2. Check: 3.2 × 3 = 9.6.
r2_s1	Set 2: Dividing by 10, 100, or 1,000 (Week 17): type the quotient. 4.2 ÷ 10 =	**0.42** (also accepted: .42)	Dividing by 10 moves every digit one place to the right (to smaller place values): 4.2 ÷ 10 = 0.42.

ID	Question	Answer	Why (explain to student)
r1_s2	Set 3: Dividing a decimal by a whole number (Week 17): type the quotient. 7.5 ÷ 5 =	**1.5**	7.5 ÷ 5: 7.5 is 75 tenths, and 75 ÷ 5 = 15 tenths, which is 1.5. Check: 1.5 × 5 = 7.5.
r2_s2	Set 3: Dividing by 10, 100, or 1,000 (Week 17): type the quotient. 250 ÷ 1,000 =	**0.25** (also accepted: .25, 0.250)	Dividing by 1,000 moves every digit three places to the right (to smaller place values): 250 ÷ 1,000 = 0.250, which is 0.25.

WARM-UP: ESTIMATE AND SOLVE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r3_s0	Set 1: Estimate, then choose (Week 17): which is the correct quotient for 16.2 ÷ 6? [2.7 / 27 / 0.27 / 2.2]	**2.7**	Estimate first: 12 ÷ 6 = 2 and 18 ÷ 6 = 3, so the answer is between 2 and 3. 16.2 ÷ 6 = 2.7. Check: 2.7 × 6 = 16.2.
r4_s0	Set 1: Money and measurement (Week 17): Four surveyors share the cost of a box of wooden marking stakes. The box costs $23.80. How much does each surveyor pay? Dollars:	**5.95** (also accepted: $5.95, 5.95 dollars)	$23.80 ÷ 4 = $5.95. Check: $5.95 × 4 = $23.80.
r3_s1	Set 2: Estimate, then choose (Week 17): which is the correct quotient for 15.6 ÷ 4? [3.9 / 39 / 0.39 / 3.4]	**3.9**	Estimate first: 12 ÷ 4 = 3 and 16 ÷ 4 = 4, so the answer is between 3 and 4. 15.6 ÷ 4 = 3.9. Check: 3.9 × 4 = 15.6.
r4_s1	Set 2: Money and measurement (Week 17): Imani and two friends split the cost of a city map book equally. The book costs $14.70. How much does each of the 3 pay? Dollars:	**4.90** (also accepted: 4.9, $4.90, $4.9, 4.90 dollars)	$14.70 ÷ 3 = $4.90. Check: $4.90 × 3 = $14.70.
r3_s2	Set 3: Estimate, then choose (Week 17): which is the correct quotient for 18.9 ÷ 9? [2.1 / 21 / 0.21 / 2.9]	**2.1**	Estimate first: 18 ÷ 9 = 2, so the answer must be close to 2. 18.9 ÷ 9 = 2.1. Check: 2.1 × 9 = 18.9.
r4_s2	Set 3: Money and measurement (Week 17): Mr. Adeyemi cuts 6.48 meters of chalk line into 4 equal pieces. How long is each piece? Meters:	**1.62** (also accepted: 1.62 meters, 1.62 meter, 1.62 m)	6.48 ÷ 4 = 1.62 meters. Check: 1.62 × 4 = 6.48.

YOUR TURN: LINES AND RAYS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Is this figure a line, a line segment, or a ray? Type your answer. Figure:	**ray** (also accepted: a ray, ray JK, JK)	It has one endpoint (J) and an arrow going on forever through K, so it is a ray. Its name is ray JK, endpoint first.
l2_s0	Set 1: Look at the map lines. Which pair of lines is perpendicular? [a and c / a and b / b and d / a and d]	**a and c**	Line c crosses line a at a square corner (a right angle), so a and c are perpendicular. Lines a and b are parallel: they never meet. Line d crosses a and b, but not at right angles.
l1_s1	Set 2: Is this figure a line, a line segment, or a ray? Type your answer. Figure:	**line segment** (also accepted: segment, a line segment, a segment, line segment MN, segment MN, MN)	It has two endpoints (M and N) and no arrows, so it is a line segment. It has a length you can measure.

ID	Question	Answer	Why (explain to student)
l2_s1	Set 2: Look at the map lines. Which pair of lines is parallel? [p and r / p and q / q and s / r and s]	**p and r**	Lines p and r slant at exactly the same steepness and never meet, so they are parallel. Line q crosses p and r, and line s crosses q, p, and r, so none of the other pairs are parallel.
l1_s2	Set 3: Is this figure a line, a line segment, or a ray? Type your answer. Figure:	**line** (also accepted: a line, line ST, ST, line TS, TS)	It has arrows on both ends and no endpoints, so it goes on forever in both directions. It is a line.
l2_s2	Set 3: Which statement is true? [Perpendicular lines meet to form right angles / Parallel lines meet at exactly one point / A ray has two endpoints / A line segment goes on forever in both directions]	**Perpendicular lines meet to form right angles**	Perpendicular lines meet to form right angles (90°). Parallel lines never meet, a ray has only one endpoint, and a line segment has two endpoints, so it does not go on forever.

YOUR TURN: COUNTING RIGHT ANGLES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l3_s0	Set 1: Two steps: Two parallel streets run east and west. A third street crosses both of them, and it is perpendicular to both. How many right angles are formed in all? Right angles:	**8** (also accepted: 8 right angles, eight)	Each crossing makes 4 right angles. There are 2 crossings, so 2 × 4 = 8 right angles.
l3_s1	Set 2: Two steps: Three parallel streets run east and west. One street crosses all three, and it is perpendicular to each of them. How many right angles are formed in all? Right angles:	**12** (also accepted: 12 right angles, twelve)	Each crossing makes 4 right angles. There are 3 crossings, so 3 × 4 = 12 right angles.
l3_s2	Set 3: Two steps: Two parallel streets run north and south. Two other parallel streets run east and west and cross them. Every crossing is perpendicular. How many right angles are formed in all? Right angles:	**16** (also accepted: 16 right angles, sixteen)	There are 2 × 2 = 4 crossings, and each crossing makes 4 right angles, so 4 × 4 = 16 right angles.

YOUR TURN: MEASURE AND SORT ANGLES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Read the protractor. Find the 0 that the base ray passes through, then type the angle's measure in degrees. Degrees:	**70** (also accepted: 70°, 70 °, 70 degrees, 70 degree, 70 deg)	The base ray points right, through the red 0, so read the red (inner) scale: the other ray crosses at 70. The angle is acute, so 70° makes sense; 110° (the blue number) would be obtuse.
k2_s0	Set 1: Which list shows only obtuse angles? [95°, 120°, 179° / 90°, 120°, 150° / 45°, 100°, 135° / 120°, 160°, 180°]	**95°, 120°, 179°**	Obtuse angles are more than 90° and less than 180°. 95°, 120°, and 179° all fit. 90° is a right angle, 45° is acute, and 180° is a straight angle.
k1_s1	Set 2: Read the protractor. Find the 0 that the base ray passes through, then type the angle's measure in degrees. Degrees:	**135** (also accepted: 135°, 135 °, 135 degrees, 135 degree, 135 deg)	The base ray points left, through the blue 0, so read the blue (outer) scale: the other ray crosses at 135. The angle is wider than a square corner, so it is obtuse, and 135° makes sense.

ID	Question	Answer	Why (explain to student)
k2_s1	Set 2: Which list shows only acute angles? [15°, 60°, 89° / 30°, 45°, 90° / 10°, 20°, 91° / 45°, 75°, 100°]	**15°, 60°, 89°**	Acute angles are less than 90°. 15°, 60°, and 89° all fit. 90° is a right angle, and 91° and 100° are obtuse.
k1_s2	Set 3: Read the protractor. Find the 0 that the base ray passes through, then type the angle's measure in degrees. Degrees:	**40** (also accepted: 40°, 40 °, 40 degrees, 40 degree, 40 deg)	The base ray points left, through the blue 0, so read the blue (outer) scale: the other ray crosses at 40. The angle is narrow (acute), so 40° makes sense; 140° (the red number) would be obtuse.
k2_s2	Set 3: Which explanation is correct? Imani says an angle that turns through 1/3 of a full circle is obtuse. [She is right: 1/3 of 360° is 120°, which is between 90° and 180° / She is wrong: 1/3 of 180° is 60°, so it is acute / She is wrong: 1/3 of a circle is 30°, so it is acute / She is wrong: 1/3 of a circle is always a right angle]	**She is right: 1/3 of 360° is 120°, which is between 90° and 180°**	A full turn is 360°, and 1/3 of 360° is 360 ÷ 3 = 120°. 120° is more than 90° and less than 180°, so the angle is obtuse. (60° comes from using 180° by mistake.)

YOUR TURN: CLOCK ANGLES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k3_s0	Set 1: Two steps: Mr. Adeyemi's watch shows exactly 2:00. What is the smaller angle between the hour hand and the minute hand? Degrees:	**60** (also accepted: 60°, 60 °, 60 degrees, 60 degree, 60 deg)	From 12 to 2 is 2 spaces. Each space is 360 ÷ 12 = 30°, so 2 × 30 = 60°.
k3_s1	Set 2: Two steps: Mr. Adeyemi's watch shows exactly 5:00. What is the smaller angle between the hour hand and the minute hand? Degrees:	**150** (also accepted: 150°, 150 °, 150 degrees, 150 degree, 150 deg)	From 12 to 5 is 5 spaces. Each space is 360 ÷ 12 = 30°, so 5 × 30 = 150°.
k3_s2	Set 3: Two steps: Mr. Adeyemi's watch shows exactly 4:00. What is the smaller angle between the hour hand and the minute hand? Degrees:	**120** (also accepted: 120°, 120 °, 120 degrees, 120 degree, 120 deg)	From 12 to 4 is 4 spaces. Each space is 360 ÷ 12 = 30°, so 4 × 30 = 120°.

YOUR TURN: ANGLE ADDITION (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Ray BC splits angle ABD into two parts. What is the measure of angle ABD? Angle ABD:	**75** (also accepted: 75°, 75 °, 75 degrees, 75 degree, 75 deg)	Angle addition: the whole angle is the sum of its parts. 25° + 50° = 75°.
e2_s0	Set 1: The two angles sit on a straight line. Find the missing angle. Missing angle:	**52** (also accepted: 52°, 52 °, 52 degrees, 52 degree, 52 deg)	The two angles make a straight angle, which is 180°. 180 − 128 = 52°.
e1_s1	Set 2: Ray BC splits angle ABD into two parts. What is the measure of angle ABD? Angle ABD:	**112** (also accepted: 112°, 112 °, 112 degrees, 112 degree, 112 deg)	Angle addition: the whole angle is the sum of its parts. 48° + 64° = 112°.
e2_s1	Set 2: The two angles sit on a straight line. Find the missing angle. Missing angle:	**143** (also accepted: 143°, 143 °, 143 degrees, 143 degree, 143 deg)	The two angles make a straight angle, which is 180°. 180 − 37 = 143°.
e1_s2	Set 3: Ray BC splits angle ABD into two parts. What is the measure of angle ABD? Angle ABD:	**105** (also accepted: 105°, 105 °, 105 degrees, 105 degree, 105 deg)	Angle addition: the whole angle is the sum of its parts. 35° + 70° = 105°.

ID	Question	Answer	Why (explain to student)
e2_s2	Set 3: The two angles sit on a straight line. Find the missing angle. Missing angle:	**105** (also accepted: 105°, 105 °, 105 degrees, 105 degree, 105 deg)	The two angles make a straight angle, which is 180°. 180 − 75 = 105°.

YOUR TURN: MORE MISSING ANGLES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Two steps: The square in the corner shows a right angle. A ray splits it into two parts. Find the missing angle. Missing angle:	**56** (also accepted: 56°, 56 °, 56 degrees, 56 degree, 56 deg)	A right angle is 90°. 90 − 34 = 56°. Check: 34 + 56 = 90.
e3_s1	Set 2: Two steps: Three angles fill the space all the way around a point. Find the missing angle. Missing angle:	**115** (also accepted: 115°, 115 °, 115 degrees, 115 degree, 115 deg)	All the way around a point is 360°. 150 + 95 = 245, and 360 − 245 = 115°. Check: 150 + 95 + 115 = 360.
e3_s2	Set 3: Two steps: The square in the corner shows a right angle. Two rays split it into three parts. Find the missing angle. Missing angle:	**25** (also accepted: 25°, 25 °, 25 degrees, 25 degree, 25 deg)	A right angle is 90°. 20 + 45 = 65, and 90 − 65 = 25°. Check: 20 + 45 + 25 = 90.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: Read the sentence from the lesson: “Put the center hole of the protractor exactly on the vertex.” What does vertex mean? [The point where the two rays of an angle meet / The number scale on the protractor / The longest side of the angle / The arrow at the end of a ray]	**The point where the two rays of an angle meet**	The vertex is the point where the two rays of an angle meet. That is where the center hole of the protractor must go.
v2_s0	Set 1: Type the missing word: two lettered streets run east and west side by side and never meet. They are ___ lines. Word:	**parallel** (also accepted: parallel lines)	Lines that run side by side and never meet are parallel lines.
v1_s1	Set 2: Read the sentence from the lesson: “A ray has one endpoint and goes on forever in one direction.” Which is a ray? [A beam of light that starts at a flashlight and keeps going / A crosswalk stripe from one curb to the other / The edge of a sheet of paper / A street that runs between two corners]	**A beam of light that starts at a flashlight and keeps going**	A ray has one endpoint and goes on forever in one direction, so it has no length you can measure. A line segment has two endpoints and does have a length.
v2_s1	Set 2: Type the missing word: a numbered street crosses the lettered streets at square corners. It is ___ to them. Word:	**perpendicular** (also accepted: perpendicular lines)	Lines that meet to form right angles (square corners) are perpendicular lines.
v1_s2	Set 3: Read the sentence from the lesson: “A 1-degree angle turns through 1/360 of a circle.” How many 1-degree angles make a full turn? [360 / 100 / 180 / 90]	**360**	A full turn is 360°, so a 1-degree angle turns through 1/360 of a circle. That is why a right angle (a quarter turn) is 90°.
v2_s2	Set 3: Type the missing word: Mr. Adeyemi measures each corner with a half-circle tool that has two number scales. The tool is called a ___. Word:	**protractor** (also accepted: a protractor)	A protractor is the half-circle tool used to measure and draw angles.

YOUR TURN: TURNS AND DRAWING (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Two steps: A cyclist turns 1/4 of a full turn at one corner, then 1/8 of a full turn at the next corner. How many degrees did she turn in all? Degrees:	**135** (also accepted: 135°, 135 °, 135 degrees, 135 degree, 135 deg)	1/4 turn = 360 ÷ 4 = 90°. 1/8 turn = 360 ÷ 8 = 45°. 90 + 45 = 135°.
t2_s0	Set 1: Which set of steps correctly draws a 40° angle? [Line up the base on the ray, then mark 40 on the scale that starts at 0 on that ray / Line up the base on the ray, then mark 40 on the scale that starts at 180 on that ray / Put the vertex at the end of the base line, not at the center hole, then mark 40 / Mark 40 on either scale; both give the same angle]	**Line up the base on the ray, then mark 40 on the scale that starts at 0 on that ray**	Use the scale that starts at 0 on your base ray. If you start from the 0 on the other side, the mark at 40 makes a 140° angle instead.
t1_s1	Set 2: Two steps: A street sweeper turns 1/6 of a full turn, then 1/3 of a full turn in the same direction. How many degrees did it turn in all? Degrees:	**180** (also accepted: 180°, 180 °, 180 degrees, 180 degree, 180 deg)	1/6 turn = 360 ÷ 6 = 60°. 1/3 turn = 360 ÷ 3 = 120°. 60 + 120 = 180°, a half turn.
t2_s1	Set 2: Imani meant to draw a 55° angle, but she made her mark at 55 on the wrong scale. What angle did she actually draw? [125° / 35° / 145° / 55°]	**125°**	The two scales add to 180 at every mark. Reading the wrong scale at the 55 mark gives the angle 180 − 55 = 125°.
t1_s2	Set 3: Two steps: A traffic camera turns 1/12 of a full turn each time it moves. It moves 5 times in the same direction. How many degrees did it turn in all? Degrees:	**150** (also accepted: 150°, 150 °, 150 degrees, 150 degree, 150 deg)	1/12 turn = 360 ÷ 12 = 30°. 5 moves: 5 × 30 = 150°.
t2_s2	Set 3: Imani draws a 160° angle. Which description should match her drawing? [Much wider than a square corner, almost a straight line / A little smaller than a square corner / Exactly a square corner / Narrow, like a slice of pizza]	**Much wider than a square corner, almost a straight line**	160° is obtuse: more than 90° and only 20° less than a straight angle (180°). So it should look wide, almost like a straight line.

REAL WORLD: MS. RUIZ'S BIKE LANES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Ms. Ruiz is planning a bike lane where a diagonal avenue crosses a straight east–west street. On the north side of the street, the avenue makes a 42° angle with the street. A new signpost line splits the other angle on the north side into two equal angles.* Two steps: what does each of the two equal angles measure? Degrees:	**69** (also accepted: 69°, 69 °, 69 degrees, 69 degree, 69 deg)	The two angles on the north side make a straight angle: 180 − 42 = 138°. Split into two equal parts: 138 ÷ 2 = 69°.
w1_s1	Set 2: *The Situation Ms. Ruiz is designing a small plaza where three walking paths meet at a fountain. The three angles between the paths fill the whole circle around the fountain. Two of the angles measure 105° and 125°.* Two steps: what does the third angle measure? Degrees:	**130** (also accepted: 130°, 130 °, 130 degrees, 130 degree, 130 deg)	All the way around the fountain is 360°. 105 + 125 = 230, and 360 − 230 = 130°.

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: *The Situation A crosswalk meets a curb at a right angle. Ms. Ruiz draws a diagonal ramp line from the corner that splits the right angle into two parts. One part is twice as large as the other.* Two steps: what does the larger part measure? Degrees:	**60** (also accepted: 60°, 60 °, 60 degrees, 60 degree, 60 deg)	Think of the right angle as 3 equal shares: the small part is 1 share and the large part is 2 shares. 90 ÷ 3 = 30°, so the larger part is 2 × 30 = 60°. Check: 30 + 60 = 90.

REAL WORLD: THE SURVEYING TOOL (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Mr. Adeyemi's surveying tool sits on a three-legged stand and turns in equal clicks. Each click turns it exactly 15°. He starts out pointing at a lamppost and turns in one direction until he points at a fire hydrant. The angle between the lamppost and the hydrant is 105°.* How many clicks does he turn? Clicks:	**7** (also accepted: 7 clicks, seven)	Each click is 15°. 105 ÷ 15 = 7 clicks. Check: 7 × 15 = 105.
w2_s1	Set 2: *The Situation Mr. Adeyemi's surveying tool sits on a three-legged stand and turns in equal clicks. Each click turns it exactly 12°. He turns it 9 clicks to the right, then 3 clicks back to the left.* Two steps: how many degrees is the tool now turned from where it started? Degrees:	**72** (also accepted: 72°, 72 °, 72 degrees, 72 degree, 72 deg)	9 − 3 = 6 clicks to the right in the end. 6 × 12 = 72°. (Or: 9 × 12 = 108 and 3 × 12 = 36, and 108 − 36 = 72.)
w2_s2	Set 3: *The Situation Mr. Adeyemi's surveying tool sits on a three-legged stand and turns in equal clicks. Each click turns it exactly 20°. Turning in one direction from his starting line, the first street sign is 5 clicks away and the second street sign is 11 clicks away.* Two steps: what is the angle between the two street signs? Degrees:	**120** (also accepted: 120°, 120 °, 120 degrees, 120 degree, 120 deg)	The signs are 11 − 5 = 6 clicks apart. 6 × 20 = 120°. (Or: 11 × 20 = 220 and 5 × 20 = 100, and 220 − 100 = 120.)

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? A ray splits a straight angle into two angles. One angle is 65°. What is the other? [115°, because angles on a straight line add to 180° and 180 − 65 = 115 / 25°, because 90 − 65 = 25 / 295°, because 360 − 65 = 295 / 65°, because the two angles must be equal]	**115°, because angles on a straight line add to 180° and 180 − 65 = 115**	The two angles make a straight angle, and a straight angle is 180°. 180 − 65 = 115°. (25° comes from using 90° by mistake, and 295° from using 360°.)
x2_s0	Set 1: Find the mistake: Callum measures an acute angle with a protractor and writes 150°. What went wrong? [He read the wrong scale. An acute angle is less than 90°, so it measures 30° / Nothing. Acute angles can be 150° / He should add 90, so the angle is 240° / He should have used a ruler, not a protractor]	**He read the wrong scale. An acute angle is less than 90°, so it measures 30°**	An acute angle is less than 90°, so 150° cannot be right. He read the wrong scale: the two scales add to 180 at every mark, so the angle is 180 − 150 = 30°.

ID	Question	Answer	Why (explain to student)
x1_s1	Set 2: Which explanation is correct? Angle PQR is split by ray QS into angle PQS = 38° and angle SQR = 47°. What is angle PQR? [85°, because the parts add up to the whole angle: 38 + 47 = 85 / 9°, because 47 − 38 = 9 / 95°, because 180 − 85 = 95 / You cannot tell without measuring with a protractor]	**85°, because the parts add up to the whole angle: 38 + 47 = 85**	Angle addition: when an angle is split into parts that do not overlap, the whole is the sum of the parts. 38 + 47 = 85°.
x2_s1	Set 2: Find the mistake: Ingrid says two streets that meet at a 90° corner are parallel because they are both straight. What is wrong? [They meet at a right angle, so they are perpendicular. Parallel lines never meet / Nothing. Straight streets are always parallel / They are parallel only if one street is longer / They are rays, so they cannot be parallel or perpendicular]	**They meet at a right angle, so they are perpendicular. Parallel lines never meet**	Parallel lines never meet. Two streets that meet at a 90° corner are perpendicular, not parallel. Being straight does not make lines parallel.
x1_s2	Set 3: Which explanation is correct? Imani asks, “Is a ray longer than a line segment?” [A ray goes on forever in one direction, so it has no measurable length; a line segment has two endpoints and a length / Yes, a ray is always exactly twice as long as a segment / No, a ray is shorter because it has only one endpoint / They are always the same length]	**A ray goes on forever in one direction, so it has no measurable length; a line segment has two endpoints and a length**	A ray goes on forever in one direction, so it has no length to measure. A line segment has two endpoints, so it has a length. Comparing their lengths does not make sense.
x2_s2	Set 3: Find the mistake: a right angle is split into two parts. One part is 27°. Corwin writes 180 − 27 = 153° for the other part. What did he do wrong? [He used 180 instead of 90. A right angle is 90°, so the part is 90 − 27 = 63° / Nothing. 153° is correct / He should add: 90 + 27 = 117° / He should use 360: 360 − 27 = 333°]	**He used 180 instead of 90. A right angle is 90°, so the part is 90 − 27 = 63°**	A right angle is 90°, not 180°. The missing part is 90 − 27 = 63°. Check: 27 + 63 = 90.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: imani asked, is a ray longer than a line	**Imani asked, "Is a ray longer than a line?"**	Capitalize the name Imani, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The protractor has it's center hole on the vertex.	**The protractor has its center hole on the vertex.**	Its (no apostrophe) shows ownership: the center hole belongs to the protractor. It's always means it is.
s3_s0	Set 1: Fix this sentence: Two lines can be parallel perpendicular or intersecting.	**Two lines can be parallel, perpendicular, or intersecting.** (also accepted: Two lines can be parallel, perpendicular or intersecting.)	Use commas to separate the items in a series: parallel, perpendicular, or intersecting.

ID	Question	Answer	Why (explain to student)
s4_s0	Set 1: Fix this sentence: the two avenues meets at an acute angle	**The two avenues meet at an acute angle.** (also accepted: The two avenues met at an acute angle.)	Capitalize the first word. The subject is avenues (plural), so use meet, not meets. End with a period.
s1_s1	Set 2: Fix this sentence: mr. adeyemi said, a right angle measures 90 degrees	**Mr. Adeyemi said, "A right angle measures 90 degrees."**	Capitalize Mr. Adeyemi (a title with a name), put quotation marks around his exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: Its easy to spot a right angle at a street corner.	**It's easy to spot a right angle at a street corner.**	It's means it is: it is easy to spot a right angle. Use the apostrophe.
s3_s1	Set 2: Fix this sentence: An angle can be acute right obtuse or straight.	**An angle can be acute, right, obtuse, or straight.** (also accepted: An angle can be acute, right, obtuse or straight.)	Use commas to separate the items in a series: acute, right, obtuse, or straight.
s4_s1	Set 2: Fix this sentence: each of the streets run north and south	**Each of the streets runs north and south.** (also accepted: Each of the streets ran north and south.)	Capitalize the first word. The subject is each (one at a time, so singular), not streets, so use runs, not run. End with a period.
s1_s2	Set 3: Fix this sentence: ms. ruiz asked, which streets are perpendicular	**Ms. Ruiz asked, "Which streets are perpendicular?"**	Capitalize Ms. Ruiz, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Every angle has it's own vertex.	**Every angle has its own vertex.**	Its (no apostrophe) shows ownership: the vertex belongs to the angle. It's means it is.
s3_s2	Set 3: Fix this sentence: Imani measured angles of 30 degrees 90 degrees and 145 degrees.	**Imani measured angles of 30 degrees, 90 degrees, and 145 degrees.** (also accepted: Imani measured angles of 30 degrees, 90 degrees and 145 degrees.)	Use commas to separate the items in a series: 30 degrees, 90 degrees, and 145 degrees.
s4_s2	Set 3: Fix this sentence: the rays of a straight angle points in opposite directions	**The rays of a straight angle point in opposite directions.** (also accepted: The rays of a straight angle pointed in opposite directions.)	Capitalize the first word. The subject is rays (plural), not angle, so use point, not points. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match acute angle	**less than 90°**	2 points if matched on the first try.
match_s0_2	Set 1: Match right angle	**exactly 90°**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s0_3	Set 1: Match obtuse angle	**between 90° and 180°**	2 points if matched on the first try.
match_s0_4	Set 1: Match straight angle	**exactly 180°**	2 points if matched on the first try.
match_s0_5	Set 1: Match full turn	**360°**	2 points if matched on the first try.
match_s1_1	Set 2: Match ray	**1 endpoint**	2 points if matched on the first try.
match_s1_2	Set 2: Match line segment	**2 endpoints**	2 points if matched on the first try.
match_s1_3	Set 2: Match line	**no endpoints**	2 points if matched on the first try.
match_s1_4	Set 2: Match parallel lines	**never meet**	2 points if matched on the first try.
match_s1_5	Set 2: Match perpendicular lines	**meet at right angles**	2 points if matched on the first try.
match_s2_1	Set 3: Match 1/4 turn	**90°**	2 points if matched on the first try.
match_s2_2	Set 3: Match 1/2 turn	**180°**	2 points if matched on the first try.
match_s2_3	Set 3: Match 1/3 turn	**120°**	2 points if matched on the first try.
match_s2_4	Set 3: Match 1/6 turn	**60°**	2 points if matched on the first try.
match_s2_5	Set 3: Match 1/12 turn	**30°**	2 points if matched on the first try.

BOSS BATTLE: THE BIG INTERSECTION (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Imani and Mr. Adeyemi Map the Big Intersection At one corner, a north–south numbered street crosses a diagonal avenue. The avenue makes a 38° angle with the north part of the street. Ms. Ruiz wants a crosswalk line through the same corner that is perpendicular to the numbered street. Later, Imani and Mr. Adeyemi map a traffic circle where 6 straight streets leave the circle, evenly spaced all the way around. 38°?numbered street (north)crosswalkavenue* Two steps: what is the angle between the crosswalk and the avenue (the angle marked?)? [52° / 38° / 128° / 142°]	**52°**	The crosswalk is perpendicular to the street, so the angle from the north part of the street to the crosswalk is 90°. The avenue takes 38° of that right angle, so the angle marked? is 90 − 38 = 52°.
b2_s0	Set 1: Two steps: in the traffic circle, what is the angle between two streets that have exactly one street between them? Degrees:	**120** (also accepted: 120°, 120 °, 120 degrees, 120 degree, 120 deg)	The 6 streets split 360° into 6 equal angles: 360 ÷ 6 = 60°. Two streets with one street between them are 2 spaces apart: 2 × 60 = 120°.
b1_s1	Set 2: *Imani and Mr. Adeyemi Map the Big Intersection A diagonal avenue crosses an east–west street and makes a 56° angle with it. A bike path starts at the same corner and makes a right angle with the avenue, inside the wide angle on that side of the street. Later, Imani and Mr. Adeyemi map a traffic circle where 8 straight streets leave the circle, evenly spaced all the way around. 56°? streetavenuebike path* Two steps: what is the angle between the bike path and the street (the angle marked?)? [34° / 124° / 56° / 44°]	**34°**	The two angles on one side of the street make 180°, so the wide angle is 180 − 56 = 124°. The bike path takes 90° of it, so the angle marked? is 124 − 90 = 34°.

ID	Question	Answer	Why (explain to student)
b2_s1	Set 2: Two steps: in the traffic circle, what is the angle between two streets that have exactly two streets between them? Degrees:	**135** (also accepted: 135°, 135 °, 135 degrees, 135 degree, 135 deg)	The 8 streets split 360° into 8 equal angles: 360 ÷ 8 = 45°. Two streets with two streets between them are 3 spaces apart: 3 × 45 = 135°.
b1_s2	Set 3: *Imani and Mr. Adeyemi Map the Big Intersection Two diagonal paths start at the same point on a straight curb and split the straight angle into three angles. The first angle measures 47° and the second measures 68°. Mr. Adeyemi's surveying tool turns exactly 9° with each click. 47°68°?* Two steps: what does the third angle measure? [65° / 55° / 113° / 245°]	**65°**	The three angles make a straight angle, 180°. 47 + 68 = 115, and 180 − 115 = 65°. (245° comes from using 360°.)
b2_s2	Set 3: Two steps: Mr. Adeyemi turns the tool from one end of the curb all the way to the other end (a straight angle). Then he turns it back 5 clicks. How many degrees is the tool now turned from where it started? Degrees:	**135** (also accepted: 135°, 135 °, 135 degrees, 135 degree, 135 deg)	A straight angle is 180°. Turning back 5 clicks is 5 × 9 = 45°. 180 − 45 = 135°.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W18.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 8.4 ÷ 4 and 52.6 ÷ 10	a) 2.1 b) 5.26
2	Draw and label a line, a segment and a ray	Line: arrows on both ends. Segment: two endpoints, no arrows. Ray: one endpoint and one arrow.
3	Rails that never cross	parallel
4	Angle smaller than a square corner	acute
5	Degrees in a quarter turn	90°
6	Draw a right angle	Drawing shows a square corner marked with a small square.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Classify 47°, 90°, 135°, 180°, 89°	a) acute b) right c) obtuse d) straight e) acute
2	Straight angle split into 38°, 75° and ?	180° − 38° − 75° = 67°. It is acute.
3	Smaller angle between the hands at 10:00	60° (2 spaces × 30°).
4	Tobin wrote 40° for an obtuse angle	An obtuse angle is more than 90°, so 40° cannot be right. He read the wrong protractor scale; the angle measures 140° (180° − 40°).

★ CHALLENGE

#	Prompt	Model answer / grading note
1	1/4 turn, then 1/12 turn in the same direction	90° + 30° = 120°. That total is obtuse.